

Advanced (Carolina Reaper)

- Q1.** What is the slope of the line $y = 2x + 3$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q2.** The slope of a line is $\frac{3}{4}$. What is the rate of change?
(A) $\frac{3}{4}$ units per hour
(B) 3 units per 4 hours
(C) $\frac{4}{3}$ units per hour
(D) 4 units per 3 hours
(E) $\frac{1}{2}$ units per hour
- Q3.** A line passes through (2,3) and (4,5). What is the slope?
(A) 0
(B) $\frac{1}{2}$
(C) 1
(D) $\frac{3}{2}$
(E) 2
- Q4.** What is the slope of the line $x = 2$?
(A) 0
(B) 1
(C) 2
(D) undefined
(E) None of the above
- Q5.** If the slope of a line is -2, what is the rate of change?
(A) -2 units per hour
(B) 2 units per hour
(C) -1 unit per 2 hours
(D) 1 unit per 2 hours
(E) -1 unit per hour
- Q6.** A line has a slope of 0. What is the rate of change?
(A) no change
(B) 1 unit per hour
(C) 2 units per hour
(D) 0 units per hour
(E) None of the above
- Q7.** The graph of a line shows a slope of $\frac{2}{3}$. What is the rate of change?
(A) $\frac{2}{3}$ units per hour
(B) 2 units per 3 hours
(C) $\frac{3}{2}$ units per hour
(D) 3 units per 2 hours
(E) $\frac{1}{2}$ units per hour
- Q8.** What is the slope of a horizontal line?
(A) 1
(B) 0
(C) -1
(D) undefined
(E) None of the above

Open Ended Questions (FRQ)

- Q9.** Determine the slope of the line that passes through the points (1,2) and (3,5) and explain the rate of change.
- Q10.** A water tank is being filled at a rate of $\frac{2}{3}$ gallons per minute. Write an equation for the amount of water in the tank after x minutes and determine the slope. Explain the meaning of the slope in this context.

Chef's Tips

- Read the questions carefully and choose the correct answer.
- Use the space provided for the free response questions.
- Calculators are not allowed.